Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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ABSTRACT

Large pre-trained language models store factual knowledge in their parameters, but their ability to access and manipulate knowledge is limited, and they prone to hallucinations. We present Retrieval-Augmented Generation (RAG) models, which combine pre-trained parametric memory (seq2seq generator) with non-parametric memory (dense vector index of Wikipedia) accessed via Dense Passage Retrieval

1. Introduction: Parametric vs Non-Parametric Memory

Pre-trained neural language models have achieved remarkable success on open-domain NLP benchmarks. However, they struggle to update their factual knowledge dynamically, can hallucinate incorrect assertions with high confidence, and cannot readily cite specific external evidence supporting their claims.

We introduce RAG, a general-purpose fine-tuning recipe for sequence-to-sequence models that augments parametric memory (BART generator) with a non-parametric retrieval component based on Dense Passage Retrieval (DPR) over a dense vector index of 21 million Wikipedia passages.

2. RAG Architectures: RAG-Sequence vs RAG-Token

The retriever p_eta(z|x) uses DPR: dense bi-encoder with query encoder q(x) and document encoder d(z) computing similarity via dot product. Top-k retrieved passages z are passed to the generator p_theta(y|x, z).

RAG-Sequence Model: Uses the same retrieved document to predict all target tokens: p_RAG-Seq(y|x) = sum_{z} p_eta(z|x) * prod_{i} p_theta(y_i | x, z, y_{1:i-1}). It generates complete candidate sequences per document and marginalizes.

RAG-Token Model: Allows each token to attend to a different retrieved document: p_RAG-Token(y|x) = prod_{i} sum_{z} p_eta(z|x) * p_theta(y_i | x, z, y_{1:i-1}). This allows the generator to synthesize information from multiple distinct documents when constructing an answer.

3. Benchmark Results, Open-Domain QA & Limitations

We evaluate RAG on open-domain question answering datasets: Natural Questions (NQ), TriviaQA, WebQuestions, and CuratedTREC, alongside Jeopardy question generation and fact verification (FEVER).

On Natural Questions, RAG achieves 44.5 Exact Match (EM), substantially outperforming closed-book T5-11B (34.5 EM) and DPR alone (41.5 EM). On TriviaQA, RAG achieves 56.1 EM.

Key Findings & Limitations: Non-parametric memory allows updating knowledge simply by swapping the passage index without retraining model weights. However, limitations include: retrieval errors propagate directly to the generator; fixed passage chunking (100 words) can fragment multi-sentence reasoning; and DPR retriever was kept frozen during generator training.